

An implied 2-jet that is not 1-implied

Candidate full counterexample to Remark 2.11 of arXiv:2210.08687

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Status: candidate full counterexample; likely valid; human review needed. In the truncated 2-jet ring $\mathcal{P}_0^2(\mathbb{R}^3)$, the ideal $I = \text{span}\{xz, yz, x^2 + y^2\}$ implies $p = x^2 + 2y^2$, but no single element of I can imply p in the sense of Remark 2.11.

Source question

Fefferman and Shaviv define a jet p to be *implied* by an ideal I when, near the allowed directions $\Omega = \text{Allow}(I)$ and at every sufficiently small scale, p is an arbitrarily small C^m error plus a tame C^m -coefficient combination of finitely many fixed elements of I . Remark 2.11 asks whether the finite list can always be replaced by a single fixed element of I ; equivalently, whether every implied jet is 1-implied [1, p. 12]. The exact definition and question are reproduced below.

Definition 2.10 (implied jets). Let $I \triangleleft \mathcal{P}_0^m(\mathbb{R}^n)$ be an ideal, denote $\Omega := \text{Allow}(I)$ and let $p \in \mathcal{P}_0^m(\mathbb{R}^n)$ be some polynomial. We say that I *implies* p (or that p is implied by I) if there exist a constant $A > 0$ and $Q_1, Q_2, \dots, Q_L \in I$ such that the following holds:

For any $\epsilon > 0$ there exist $\delta, r > 0$ such that for any $0 < \rho \leq r$ there exist functions $F, S_1, S_2, \dots, S_L \in C^m(\text{Ann}_4(\rho))$ satisfying

$$(28) \quad |\partial^\alpha F(\vec{x})| \leq \epsilon \rho^{m-|\alpha|} \text{ for all } \vec{x} \in \text{Ann}_4(\rho) \text{ and all } |\alpha| \leq m;$$

$$(29) \quad |\partial^\alpha S_l(\vec{x})| \leq A \rho^{-|\alpha|} \text{ for all } \vec{x} \in \text{Ann}_4(\rho), \text{ all } |\alpha| \leq m \text{ and all } 1 \leq l \leq L;$$

$$(30) \quad p(\vec{x}) = F(\vec{x}) + S_1(\vec{x})Q_1(\vec{x}) + S_2(\vec{x})Q_2(\vec{x}) + \dots + S_L(\vec{x})Q_L(\vec{x})$$

for all $\vec{x} \in \text{Ann}_2(\rho)$ such that $\text{dist}(\frac{\vec{x}}{|\vec{x}|}, \Omega) < \delta$.

Remark 2.11. Definition 2.10 asserts the existence of functions $S_1, \dots, S_L$ for some $L \in \mathbb{N}$. One can define "1-implied jets" by only allowing $L = 1$, so for instance (30) becomes " $p(\vec{x}) = F(\vec{x}) + S_1(\vec{x})Q_1(\vec{x})$ for all $\vec{x} \in \text{Ann}_2(\rho)$ such that $\text{dist}(\frac{\vec{x}}{|\vec{x}|}, \Omega) < \delta$ ". Clearly if a jet is 1-implied by an ideal it is also implied by the same ideal. We do not know whether the converse also holds; in particular we are not able to construct an ideal I and a jet $p \in \mathcal{P}_0^m(\mathbb{R}^n)$, such that p is implied by I , but p is not 1-implied by I . Similar remarks can be made regarding Definition

Result and proof intuition

All products below are products in the truncated, nonunital jet ring $\mathcal{P}_0^2(\mathbb{R}^3)$: terms of degree greater than 2 are discarded.

Theorem 1. *Let*

$$I = \text{span}_{\mathbb{R}}\{xz, yz, x^2 + y^2\} \subset \mathcal{P}_0^2(\mathbb{R}^3), \quad p = x^2 + 2y^2.$$

Then I is an ideal, p is implied by I , and p is not 1-implied by I . Consequently the question in Remark 2.11 has a negative answer.

Proof intuition. The allowed directions of I are precisely the two poles of the unit sphere. Near those poles, two generators recover p by the elementary identity

$$p = \frac{x}{z}(xz) + \frac{2y}{z}(yz).$$

The apparent freedom of using two generators is essential. A single $Q \in I$ has the form $Q = axz + byz + c(x^2 + y^2)$. At a pole, a first derivative in the transverse (a, b) direction forces its coefficient S to be arbitrarily small. A second derivative in the perpendicular transverse direction then cannot reproduce the nonzero Hessian of p . When $(a, b) = 0$, subtracting the xx and yy second-derivative identities cancels S and gives the contradiction directly.

Proof

Every element of I is homogeneous of degree 2. Multiplying it by any element of the nonunital ring $\mathcal{P}_0^2(\mathbb{R}^3)$ produces only terms of degree at least 3, hence zero after truncation. Thus I is an ideal. Also $p \notin I$: in a representation of p by the displayed generators, the coefficient of $x^2 + y^2$ would have to be both 1 (from x^2) and 2 (from y^2).

We first determine the allowed directions. If $\omega = (\omega_x, \omega_y, \omega_z) \in S^2$ is not a pole, then $\omega_x^2 + \omega_y^2 > 0$. By continuity, on a conic neighborhood of ω one has

$$x^2 + y^2 \geq c|(x, y, z)|^2$$

for some $c > 0$. Since $x^2 + y^2 \in I$, this makes ω forbidden in the sense of Definition 2.6. Conversely, every element of I vanishes on the punctured z -axis. No finite family from I can therefore satisfy the required positive lower bound on a cone containing either pole. Hence

$$\Omega = \text{Allow}(I) = \{(0, 0, 1), (0, 0, -1)\}. \quad (1)$$

We next show that I implies p . Choose a smooth, degree-zero angular cutoff χ on $\mathbb{R}^3 \setminus \{0\}$ which equals 1 in conic neighborhoods of both poles and is supported where $|z| \geq \frac{1}{3}|(x, y, z)|$. On the punctured space define, with the evident smooth zero extension off the support of χ ,

$$S_1 = \chi \frac{x}{z}, \quad S_2 = 2\chi \frac{y}{z}, \quad F = 0, \quad Q_1 = xz, \quad Q_2 = yz.$$

The functions S_1, S_2 are smooth and homogeneous of degree zero. Thus, for every multi-index $|\alpha| \leq 2$, there is a fixed constant A such that on $\text{Ann}_4(\rho)$,

$$|\partial^\alpha S_j| \leq A\rho^{-|\alpha|} \quad (j = 1, 2).$$

After reducing the angular parameter $\delta > 0$ if necessary, $\chi = 1$ whenever $\text{dist}(X/|X|, \Omega) < \delta$. There, at every scale,

$$F + S_1 Q_1 + S_2 Q_2 = \frac{x}{z}(xz) + \frac{2y}{z}(yz) = p.$$

The zero function satisfies all the ε -small bounds required of F . These fixed A, Q_1, Q_2 , with any sufficiently small radius r , verify Definition 2.10 for every $\varepsilon > 0$.

It remains to rule out 1-implication. Suppose to the contrary that a fixed $Q \in I$ and a constant A witness it. Write

$$Q = axz + byz + c(x^2 + y^2). \quad (2)$$

For every $\varepsilon > 0$, Definition 2.10 would then supply $\delta, r > 0$; for each $0 < \rho \leq r$ it would supply $F, S \in C^2(\text{Ann}_4(\rho))$ such that

$$p = F + SQ \quad (3)$$

on the open part of $\text{Ann}_2(\rho)$ whose directions are within δ of Ω , and

$$|\partial^\alpha F| \leq \varepsilon \rho^{2-|\alpha|} \quad (|\alpha| \leq 2). \quad (4)$$

Fix such a ρ and evaluate derivatives at the pole point $v = (0, 0, \rho)$, an interior point of the equality region.

First suppose $d = (a^2 + b^2)^{1/2} > 0$. Let

$$e = (a/d, b/d, 0), \quad t = (-b/d, a/d, 0).$$

At v one has

$$Q = 0, \quad D_e Q = \rho d, \quad D_t Q = 0, \quad D_t^2 Q = 2c.$$

Since $D_e p(v) = 0$, differentiating (3) in the e direction and using (4) gives

$$|S(v)|\rho d = |D_e F(v)| \leq \sqrt{2}\varepsilon\rho, \quad |S(v)| \leq \frac{\sqrt{2}\varepsilon}{d}. \quad (5)$$

Differentiating twice in the t direction eliminates all product-rule terms except $S(v)D_t^2 Q(v)$, so

$$D_t^2 p(v) = D_t^2 F(v) + 2cS(v).$$

Here $D_t^2 p = 2t_x^2 + 4t_y^2 \geq 2$, while the componentwise bounds in (4) imply $|D_t^2 F(v)| \leq 2\varepsilon$. Consequently

$$2 \leq 2\varepsilon + \frac{2\sqrt{2}|c|}{d}\varepsilon,$$

which is impossible after choosing $\varepsilon > 0$ sufficiently small.

Finally suppose $a = b = 0$, including the possibility $Q = 0$. At v , $Q = D_x Q = D_y Q = 0$, and differentiating (3) twice yields

$$2 = F_{xx}(v) + 2cS(v), \quad 4 = F_{yy}(v) + 2cS(v).$$

Subtracting gives $2 = F_{yy}(v) - F_{xx}(v)$, whereas (4) makes the absolute value of the right-hand side at most 2ε . Taking $\varepsilon < 1$ is the final contradiction. Thus no $Q \in I$ can witness 1-implication, and the theorem follows. $\square$

Scope and novelty note

The example answers exactly the implication-versus-1-implication question in Remark 2.11; it does not address the analogous variants mentioned there for Definitions 3.11 and 3.13. The run's four cheap indexes and targeted searches through 11 August 2026 for the exact terminology, the paper title, and follow-up work found the source paper and its sequel but no resolution of this question. Novelty is therefore provisional pending expert literature review.

References

- [1] C. Fefferman and A. Shaviv, *A property of ideals of jets of functions vanishing on a set*, arXiv:2210.08687 (2022).